



MSc Cyber Security Engineering

Digital Forensics Investigation

Digital Forensics Investigation Report

Submitted By:
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I hereby declare that this assessment is my own work and that all sources have been acknowledged.

2025-26

Title of the action: Rex vs Djanko

Court reference number: D2345X

Final report of: 5746635

Dated: 15-12-2025

Specialist field: Digital Forensics

On behalf of the: Claimant

On the instructions of: WM Police

Subject matter: Possession and supply of Jelly Bean drugs by Paul

Djanko, 22-3-23

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1. Introduction

This work was undertaken in my capacity as a digital forensics specialist in response to instructions associated with this case. On 22nd March 2023, West Midlands Police executed a warrant at the home address of Paul Djanko (“the Suspect”) following intelligence linking him to the possession and distribution of Class A, B, and C *Jelly Bean drugs* under the Misuse of Drugs Act 2007.

The purpose of this investigation is to examine the computer exhibit seized during the warrant, labelled **MMX-Djanko-PC.E01** (MD5: 608ec9b3ccb38f87aeee4770b1483c53; SHA1: c5d50594bc4bdeaa77cf49d0c741e67a04ec2a59), which was provided as a pre-ingested forensic case. The investigation aims to determine whether the suspect is connected to the possession and/or distribution of drugs, whether any other illegal activity is present, whether other suspects can be identified, and to report any additional relevant findings.

The examination was conducted following established digital forensic procedures. Detailed methodology, including the forensic tools used, artefacts examined, and workflow applied, is provided in **Appendix A1**.

2. Investigation

2.1 Investigative Findings

2.1.1 System Overview

The system under examination is a Windows desktop computer acquired from the forensic image **MMX-Djanko-PC.E01**, with local accounts including “Guest”, “Administrator”, “djanko” (Administrators group) and “PaulD” (Users group). The primary active user account during the period of interest is ‘PaulD’, created on 14 February 2023 and showing logons up to 17 February 2023, with the majority of user artefacts (emails, web history, downloads) associated with this profile. The primary system volume (vol3) is formatted with an NTFS-type file system (partition type 0x07), containing the Windows system directories and user profiles.

The system timezone was configured to GMT, and all timestamps referenced in this report are presented in GMT for consistency. The Windows installation date was identified as 12 February 2023, and the period of activity most relevant to this investigation spans 12-17 February 2023.

2.1.2 Communications

Email communications recovered from the system [A5] demonstrate coordination of Jelly Bean drops at Warwick and Birmingham universities during 14-17 February 2023.

Exchanges between Paul Djanko and Stuart Tchaikomsiva show planning and execution of three Warwick drops on 16 February 2023, with discussion of CCTV avoidance, product type ("blue beans"), pricing negotiation (rising from £30 to £40 per drop), and collection logistics. Stuart confirms completion and reports his payment share. Follow-up emails arrange two further drops for 19 and 23 February at £35 per drop. The original outbound email from Paul attaching the maps is not present in the recovered dataset; however, Stuart's reply confirms receipt of map-based instructions along with a discussion of "red beans", which Paul describes as difficult to obtain and high risk [A5].

A parallel thread between Paul Djanko and birminghamdrops@gmail.com arranges a Birmingham University drop for 16 February 2023, including collection and delivery locations, payment (£35), and a post-job cash handover. Drops Birmingham confirms completion and accepts instructions for two additional Birmingham drops on 19 and 23 February [A5].

Additional emails involve Paul and Denise ("Dinny") Djanko corresponding with social contacts (Reuben Beethohmahler, Brahm Ludwig) on non-drug matters. One email dated 17 February 2023 shows Denise requesting advice on hiding files and disk encryption [A5].

2.1.3 Images

Several image files relevant to the drop arrangements were recovered from the PaulD desktop, including *SW.png*, *3Drops16-2-23.png*, *blue.png*, *collectionBirminghamVincentDrive.png*, *dropoffbham.png*, *warwick campus.png*, *warwick drop 190223.png*, and *warwick drop 230223.png* [A4]. The files include annotated maps of the Warwick and Birmingham university areas showing collection points, routes, and sequential drop locations, together with an image of blue "beans". Filenames of several images correspond to attachments referenced in Paul's emails to Stuart and Drops Birmingham for drops on 16, 19, and 23 February 2023 (e.g., *SW.png*, *3drops.png*, and *dropoffbham.png*) [A5]. Filesystem timestamps and recent-document artefacts [A6,A7] indicate that the Warwick-labelled images were created and/or accessed on 16 February 2023, supporting that these maps were actively prepared and viewed during planning for the subsequent drops on 19 and 23 February 2023.

2.1.4 Malware and executables

I found three items of suspected malware [A9] on the system: two under the C:\Users\PaulD\Music\ folder, "microsoft office 2007 service pack 2.exe" and "mcpatcher.exe", and a separate installer "StartSetup_20221.exe" in C:\Users\PaulD\Downloads. There were additional subfolders named "Bombermania.exe", "0.exe" and "1.exe" under the Music folder that did not contain further executables . All three items were suspected to be adware or potentially unwanted software and were present on the system during the main period of Jelly Bean-related activity.

2.1.5 Windows registry

I found out that Windows 8.1 was installed on 11 February 2023, which aligns with the earliest significant user activity on the system [A6]. Additional registry timestamps indicate normal shutdown events, with the last recorded shutdown occurring at 15:35 on 17 Feb 2023 after the period of peak activity, confirming that the system remained in regular use during the timeframe of the alleged Jelly Bean-related planning.

USB storage-related key (USBSTOR) recorded that one or more removable storage devices had been connected to the system prior to seizure. While no specific Jelly Bean-related content was recovered directly from these registry entries, the presence of attached USB devices indicates potential use of external media, which may be relevant for further enquiry if associated physical devices are available for examination.

The local “djanko” account (Administrators group) was created on 11 February 2023, while the “PaulD” account (Users group) was created on 14 February 2023, with last-logon values for both accounts falling on 17 February 2023. Combined with the distribution of artefacts elsewhere on the system, these timestamps support the conclusion that “PaulD” was the main interactive user profile during the key period, with “djanko” functioning primarily as an earlier or more administrative account.

The “djanko” profile mainly opened PDF files about Windows 8 encryption and security guides, matching the browser activity [A8.1]. Entries for “PaulD” list image and document files whose names correspond to maps, a blue jelly image and screenshots used in the Warwick and Birmingham drop-planning emails, linking that profile to the Jelly Bean drop-off material.

2.1.6 Shellbags, Jumplists and Prefetch

Shellbag and Jump List [A7] artefacts for the “PaulD” profile show repeated access on 15-16 February 2023 to image, map, and PDF files whose filenames correspond to the marked maps, screenshots, and Jelly Bean imagery described in Sections 1.1.2 and 1.1.4, and attached to the Warwick and Birmingham drop-planning emails [A5]. This demonstrates that these files were not only received via email but were actively opened and reused locally during the planning of the drops.

Prefetch runs further show that “microsoft office 2007 service pack 2.exe” was executed twice on 16 February 2023 and created temporary directories containing executables such as “NORTHSTAR.EXE” and “SYSTEM.DLL”, which are not legitimate Office update components [A7]. No equivalent execution artefact was identified for “mcpatcher.exe”, while “StartSetup_20221.exe” was executed twice on the same date.

2.1.7 Browsing history

Web browsing entries [A8.1] from the primary user profile ‘PaulD’ between 16 and 17 February 2023 show activity involving Google Maps, PDF-reader downloads, and dark web research, matching the main period of interest.

On 16 February 2023, the user accessed Google Maps to view streets and campus-style locations consistent with those described in the Warwick and Birmingham drops email threads, including Squires Close, Lynchgate car park, Whitefields halls, Vincent Drive and university buildings in the Birmingham campus area. This coincides with instructions to couriers to perform drops on 16 February and supports the use of online mapping to plan those locations.

I also identified searches such as “supplier red jelly beans Birmingham” and visits to wholesale confectionery and dark-web-related information sites. In the context of the Jelly Bean drops and the separate email discussion of “red beans”, this pattern shows research into suppliers and possible online channels that could be used for illicit trade.

On 17 February 2023, the user “djanko” ran Google searches including “how to hide files on windows 8”, “how to hide files on windows 8 filetype:PDF”, “how to encrypt a partition in windows 8” and “how to encrypt a partition in windows 8 filetype:PDF”, then accessed multiple PDF guides on encryption and data hiding. This browsing activity corresponds with Denise’s email about wanting the best method of hiding files and researching hard-disk encryption.

Web history also shows visits to a shared Dropbox link associated with Reuben’s messages, which contained Batman-related images with no drug-related or hidden-data content identified [A8.1].

2.1.8 Cache and Cookies

The Chrome HTTP cache for the ‘PaulD’ profile contains multiple Google Maps tile and vector resources (mapsvt and khms tiles) for zoom levels and coordinates corresponding to Squires Close, Lynchgate car park, Whitefields halls and surrounding campus roads, as well as the Birmingham university area, on 16 February 2023 [A8.2]. These cached artefacts are consistent with the mapping activity in the web history and show that detailed route and location information was viewed around the time the Warwick and Birmingham drops were being arranged.

I identified image resources from bulkwholesalesweets.co.uk and candyroom.co.uk depicting different Jelly Bean products, including red beans, green beans and mixed-colour “Jelly Bean” packs, cached on 17 February 2023 [A8]. This closely matches the timeframe of web searches for “supplier red jelly beans Birmingham” and visits to wholesale confectionery sites, indicating that the user viewed specific product pages for red and mixed Jelly Beans rather than only generic catalogue listings.

Cookie artefacts include standard authentication and analytics cookies for Google, YouTube and Dropbox, alongside cookies set by bulkwholesalesweets.co.uk and candyroom.co.uk, such as shopifys, shopify_pay_redirect, cart_currency, stripe_mid and stripe_sid, created on 17 February 2023 [A8.3]. In combination with the confectionery images cached in [A4.3], these

cookies support that the user not only browsed Jelly Bean supplier sites but progressed to viewing or initiating checkout-related flows for Jelly Bean products.

Additional cookies from security and privacy vendors such as ExpressVPN, Avast, Kaspersky and Clario are also present in [A8.2]. Together with cache entries for dark-web (such as App Store page for a dark-web browser) and VPN-related content in [A8], these artefacts show that the user interacted with security and anonymity-tool sites during the same general period as the Jelly Bean supplier activity.

3. My opinion

3.1 Possession, supply and distribution

In my opinion, to a high degree of confidence, Paul Djanko was actively involved in organising and coordinating Jelly Bean supply and distribution during 14-17 February 2023.

Email communications from the PaulD account [A5] constitute primary evidence. These exchanges are operational, not exploratory, including specific locations, timings, payment arrangements, identification protocols, and CCTV avoidance planning. Pricing negotiation (£30 to £40 for Warwick; £35 for Birmingham) and post-execution confirmations indicate sustained coordinated activity.

Corroborating artefacts strengthen this interpretation. Google Maps browsing [A8.1] for drop locations on 16 February coincides with scheduled drops. Cached map tiles [A8.2] and locally stored annotated maps [A4] whose filenames match email attachments demonstrate preparation and distribution. Shellbag and jumplist artefacts [A7] confirm repeated file access during planning.

On the balance of probabilities, the evidence supports possession with intent to supply. Multiple drops across two campuses, use of intermediary couriers, and structured payment arrangements are inconsistent with personal possession. While no physical product was recovered, the communications, imagery, and browsing behaviour establish Paul Djanko's role as distribution coordinator.

3.2 Other illegal activity

Evidence of additional illegal activity is present but less conclusive.

On 17 February 2023, immediately after the 16 February drops, the *PaulD* user searched for "supplier red jelly beans Birmingham" and visited wholesale confectionery sites [A8.1]. Cached product images [A8.2] and Shopify/Stripe payment cookies [A8.3] indicate progression into checkout workflows. However, no order confirmations, invoices, or receipts were recovered, so completed purchases cannot be established.

Browser artefacts also show dark-web informational access and anonymity tool research on 17 February [A8.1, A8.3]. While this suggests intent to conceal future activity, no evidence demonstrates successful deployment or separate identifiable offences.

In summary, the evidence supports preparatory activity for further supply, including supplier research and anonymisation exploration, but does not establish completed procurement or additional substantive offences.

3.3 Other suspects

Stuart Tchaikomsiva and the operator of *birminghamdrops@gmail.com* ("Drops Birmingham") are active participants functioning as couriers. Both received detailed drop instructions [A5], demonstrated product knowledge (Stuart discusses "blue beans", "red beans", "class", and sentencing), confirmed campus familiarity, executed drops, and accepted payment (Stuart: £120; Drops Birmingham: £35).

Other individuals:

- "Bill" - Received cash proceeds from Warwick drops; associated with collection vehicle. Role unclear (courier/supplier/intermediary). Insufficient evidence.
- "Kevin" - Referral source for Drops Birmingham contact. No independent communications. Insufficient evidence.
- "Lez" - Mentioned briefly; context unclear. Insufficient evidence.

Denise ("Dinny") Djanko - On 17 February 2023, emailed seeking file-hiding advice [A5]. The *djanko* profile accessed encryption guides [A8.1], and two password-protected PDFs exist under her directory. However, no direct link to Jelly Bean activity was established. Browser activity shows no overlap with drop-related mapping or supplier research. While timing indicates awareness and concealment intent, the evidence does not support a finding of direct involvement in supply or distribution. She is a person of interest for potential evidence concealment.

Reuben Beethohmahler and Brahm Ludwig ("Pedro") appear only in social and image-sharing email contexts, with no references to drugs, Jelly Beans, or drop-related activity. On the evidence examined, they do not appear to be involved in Jelly Bean supply or distribution.

3.4 Other relevant issues

Limitations:

1. Missing outbound emails - Stuart's 17 February reply references Paul's "2 more for you" email with map attachments, which was not recovered.
2. Inaccessible encrypted material - Two password-protected PDFs under Denise's profile could not be examined.

3. Third-party accounts - No independent verification of email accounts or identities available from this exhibit alone.

User attribution:

The *PaulD* account is attributed to Paul Djanko and *djanko* to Denise Djanko based on: email correspondents addressing "Paul" and "Dinny" [A5]; activity nature (PaulD: Jelly Bean emails/maps/supplier research; djanko: encryption guides); account creation timing [A6]; and shellbag/jumplist patterns [A7]. While shared use cannot be entirely excluded, on the balance of probabilities, attribution to Paul and Denise respectively is sound.

Overall conclusion:

The evidence strongly supports Paul Djanko's role coordinating Jelly Bean supply and distribution during 14-17 February 2023, with Stuart Tchaikomsiva and Drops Birmingham as active couriers. Denise Djanko is a person of interest for evidence concealment but not directly implicated in distribution. Other individuals lack sufficient evidence (Bill, Kevin, Lez) or appear uninvolved (Reuben, Brahm).

3. Appendices

A1. Methodology

A1.1 Background

The examination was conducted in accordance with standard digital forensic practice using a structured, artefact-driven approach. The provided forensic image (**MMX-Djanko-PC.E01**) was first verified using **Exterro FTK Imager** to confirm its integrity prior to analysis. All subsequent examination was performed on the verified forensic copy in a read-only manner to preserve evidential integrity.

Analysis focused on identifying and correlating user activity artefacts relevant to the investigation timeframe. Key artefact categories examined included email data, web browsing history, cache and cookies, downloaded files and executables, Windows Registry hives, Shellbags, Jump Lists and Prefetch execution records. These artefacts were selected to establish user attribution, timeline reconstruction, application execution, and evidence of planning or coordination activity.

Timestamps were interpreted with reference to the system time configuration and normalised to **GMT** for consistency across artefacts. Findings were evaluated by cross-correlating independent sources to identify consistent patterns of activity and to reduce reliance on any single artefact type. Where limitations or uncertainties were identified, these are noted in the relevant sections of the report.

A1.2 Statement of methodology

A1.2.2 Overall Workflow

The investigation followed a structured, top-down workflow. Initial analysis used Autopsy's ingest results to obtain an overview of user activity, including emails, web browsing artefacts, filesystem content and executable files. Based on this overview, the examination then focused on specific artefact categories relevant to user behaviour, program execution and timeline reconstruction, including emails, shellbags, jump lists, prefetch files, registry hives and potentially malicious executables.

The investigation timeframe was narrowed to 12-17 February 2023 based on initial email review showing the first Jelly Bean-related communication on 14 February and the last relevant activity on 17 February, with 12 February included to capture system installation context. This date scoping was applied consistently across all subsequent artefact analysis to maintain focus on the relevant period and reduce noise from unrelated activity.

Where appropriate, specialist third-party tools were used to re-examine extracted artefacts to validate and extend Autopsy's findings, and results from multiple artefact sources were correlated to support consistent interpretation and chronological reconstruction.

A1.2.2 Email and communications analysis

Email artefacts were initially examined using Autopsy's built-in email and message analysis modules to identify user accounts, correspondents and relevant communication threads. Emails referencing Jelly Bean distribution activities were identified through keyword filtering using terms including "drop", "beans", "blue", "red", "Warwick", "Birmingham", "CCTV", and correspondent names "Stuart" and "Birmingham", combined with date scoping to 14-17 February 2023. These keywords were selected based on the police intelligence brief referencing university drops and Jelly Bean terminology, and were refined iteratively as additional relevant terms emerged during initial review.

Selected messages were extracted into structured tables [A5] organized by correspondent thread (Paul-Stuart, Paul-Drops Birmingham, Paul-Denise-social contacts) to support comparison across conversations and to assist with correlating communications against other system activity such as map file creation and web searches.

A1.2.3 Malware identification and YARA-based analysis

Potentially suspicious executables and folders were first identified through Autopsy's file type, metadata and directory analysis, with attention paid to non-standard binaries stored within user-accessible locations such as the Music and Downloads folders rather than Program Files. This approach was taken because legitimate software typically installs to system directories, whereas malware and potentially unwanted programs often reside in user-writable locations to avoid requiring administrator privileges.

YARA [A9] was then used as a secondary identification technique to flag files matching rule patterns associated with suspicious tools and naming conventions. Specific YARA rules targeting office-themed executables (e.g., "microsoft office" in filename but non-Microsoft signature) and generic adware patterns were applied to the identified executables. The presence and execution of these files were subsequently examined in conjunction with prefetch artefacts to assess their relevance to user activity rather than assuming malicious causation in isolation.

A1.2.4 Shellbags, jump lists and user interaction artefacts

Windows shellbag data was examined via Autopsy to identify folders accessed by the user, including paths leading to directories containing suspicious executables and bean-drop maps. Jump list files extracted from the Autopsy case were then analysed using Jumplist Explorer to determine which executables and file paths had been accessed via the Windows taskbar or Start menu.

Priority was given to shellbag and jumplist entries for the *PaulD* profile within the 14-17 February 2023 timeframe, and specifically to paths containing filenames matching those referenced in email attachments (e.g., SW.png, 3drops.png, dropoffbham.png). This targeted approach allowed verification that map and image files were not only received via email but were actively opened and reused locally during the planning period.

A1.2.5 Registry analysis

Key Windows registry hives, including DEFAULT, SYSTEM, SOFTWARE, SECURITY and SAM, were exported from the Autopsy case and examined in Registry Explorer to reconstruct system configuration and usage. Registry keys under CurrentVersion and SYSTEM were used to identify the system owner, computer name, installation date, last shutdown time and configured time zone, while USBSTOR and related keys were reviewed for evidence of USB device connections relevant to the case chronology.

UserAssist and RecentDocs keys for both *PaulD* and *djanko* profiles were specifically examined to identify recently opened documents and compare activity patterns between the two accounts. This comparison was necessary to support user attribution and to determine whether the *djanko* account showed any overlap with Jelly Bean-related file access, or whether it was limited to encryption-related activity as the email evidence suggested.

A1.2.6 Prefetch and program execution analysis

Prefetch files identified within the Autopsy case were analysed using PECmd to extract program execution timestamps, run counts and associated file paths. Analysis focused on executables identified as suspicious in section A1.2.3 (microsoft office 2007 service pack 2.exe, mcpatcher.exe, StartSetup_20221.exe) to determine whether they had been executed during the 14-17 February investigation period. Prefetch data was then correlated with email communications, shellbag entries and jump list artefacts to establish temporal relationships between user communications, file access and program execution.

A1.2.7 Cache and cookies

For web-related artefacts, Chrome browsing history, HTTP cache and cookie data for the *PaulD* and *djanko* profiles were exported into CSV format and reviewed using spreadsheet and forensic-analysis tools [A8]. Filtering was applied by date (12-17 February 2023), by relevant host names (google.com/maps, bulkwholesalesweets.co.uk, candyroom.co.uk, tor-related sites) and by keyword patterns such as "jelly", "bean", "Birmingham", "Warwick", "encrypt", "hide" and "dark web" to isolate items related to the alleged Jelly Bean activity and encryption research.

For the cache, priority was given to entries whose URLs or content-types indicated map tiles (to corroborate Google Maps usage for drop locations), product images from confectionery sites (to support supplier research), and security-tool pages (to establish anonymisation intent). Cache timestamps were correlated with browsing history to confirm that cached content was accessed during the investigation timeframe rather than persisting from earlier unrelated activity.

Cookie analysis concentrated on domains seen in the history and cache, with particular attention to Shopify and Stripe cookies on sweets-supplier domains, as these indicate progression beyond browsing into checkout or payment workflows. Cookies were examined for creation timestamps on 17 February 2023 to align with the "red beans" supplier research identified in email and browser history.

A1.2.8 Correlation and chronology construction

Following analysis of each artefact category, relevant timestamps and events were collated into a unified chronology [A2], with particular focus on the dates identified through email communications and the police brief (14, 16, 17 February 2023 as key activity dates). Cross-correlation of independent artefacts was used to validate findings and reduce reliance on any single evidence source. For example, email references to map attachments were corroborated by filesystem timestamps on corresponding image files, shellbag evidence of file access, and cached Google Maps tiles for the same geographic locations, creating a multi-artefact chain of evidence supporting the planning and execution of drops.

This correlation process formed the basis for the timeline reconstruction presented in Figure 1 and informed the opinions presented in Section 3 of the report.

A2. Chronology

Date / Time	Event Summary	Evidence Source
14 Feb 2023 19:43-20:19	Stuart initiates email contact with Paul regarding a planned Warwick University “drop”.	Emails (Paul-Stuart) [A5]
15 Feb 2023 06:46-08:50	Paul emails Stuart setting out plans for three Warwick drops on 16 Feb, including collection, drop locations, and pricing.	Emails (Paul-Stuart) [A5]
15 Feb 2023 20:12-20:29	Paul and Stuart exchange emails about payment for the Warwick drops and share location and product images.	Emails and attachments [A5,A4]
15 Feb 2023 20:37-20:57	Paul emails Drops Birmingham to arrange a Birmingham University drop for 16 Feb, providing collection and drop details.	Emails (Paul-Drops Birmingham) [A5]
15 Feb 2023 20:46-21:01	Paul emails Stuart confirming a higher per-drop rate for Warwick and sends a route map for the drops.	Emails and attachment [A5-A4]
16 Feb 2023 12:27	Stuart emails Paul confirming completion of the Warwick job and reporting his share of proceeds.	Emails (Stuart-Paul) [A5]
16 Feb 2023 14:09 - 14:56	Drops Birmingham emails Paul confirming completion of the Birmingham University drop. Paul gives two more jobs, Birmingham drops on 19 and 23.	Emails (Drops Birmingham-Paul) [A5]
17 Feb 2023 11:43	Stuart emails Paul regarding further planned Warwick drops on 19 and 23 Feb and mentions interest in “red beans”.	Emails (Stuart-Paul) [A5]
17 Feb 2023	Browser artefacts show visits to dark-web resources, a Google search for “red jelly beans Birmingham”, and visits to a jelly-bean retail site.	Web history and cookies [A8]
17 Feb 2023 14:08	Paul emails Stuart noting difficulties and risk in obtaining “red beans” and the need for an alternative supplier.	Emails (Paul-Stuart) [A5]

Table A2.1

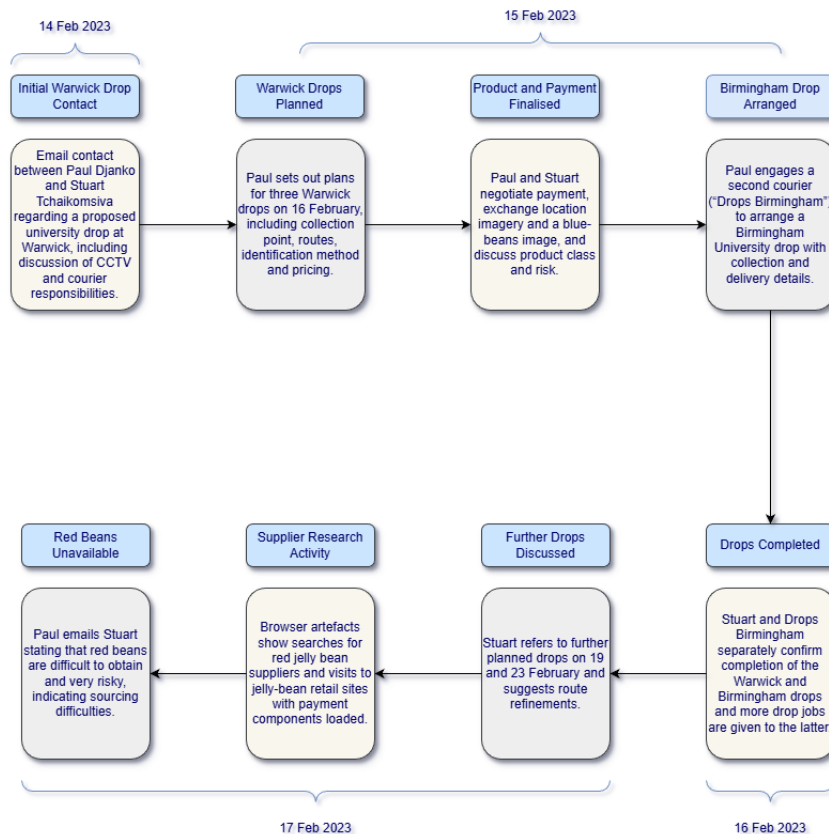


Figure 1 - Key Events Timeline: Drop Planning and Activity

A3. Glossary

Artefact - Any recoverable trace of user or system activity (such as logs, files, or registry entries) that can be interpreted as digital evidence.

Forensic image - A bit-for-bit copy of a digital storage device that preserves all data (including deleted space) for examination.

GMT (Greenwich Mean Time) - Time standard used in this report to normalise timestamps from system artefacts.

Hash value (MD5, SHA1) - Fixed-length cryptographic checksum used to verify the integrity and identity of files or forensic images.

Jump list - Windows artefact recording recently or frequently opened items for a program, used to infer file access by a user.

NTFS - New Technology File System; the main Windows file system, supporting permissions, timestamps, and other metadata used in forensics.

Prefetch file - Windows file that records information about program executions to speed up loading and can be used to infer run count and last run time.

Registry hive - Major section of the Windows Registry stored as a file on disk (for example SYSTEM, SOFTWARE, NTUSER.DAT) containing configuration and usage data.

Shellbags - Windows artefacts recording folder view settings and access history, used to infer which directories a user has browsed.

Timeline / chronology - Correlated list of events ordered by time, built from multiple artefact sources to reconstruct user activity.

YARA - Pattern-matching tool and rule language used to identify files (such as malware) based on textual or binary signatures.

A4. Images

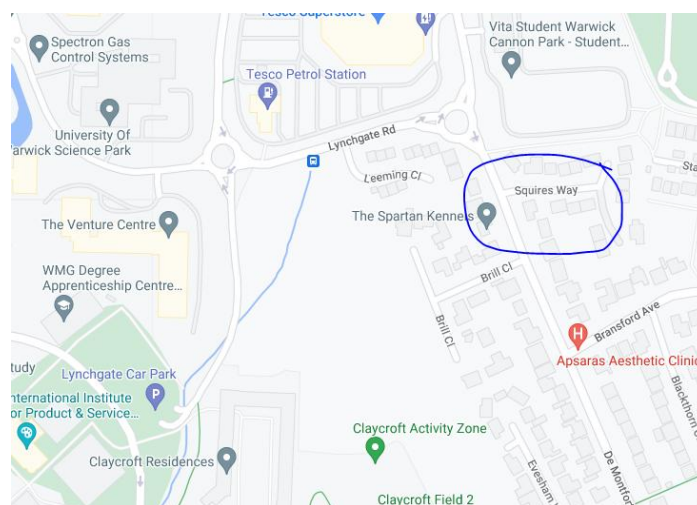


Figure A4.1 - SW.png, Annotated map showing Squires Way collection point (red Ford Focus) referenced in Paul-Stuart emails for 16 February 2023 Warwick drop, recovered from PaulD desktop [A5].



Figure A4.2 - blue.png, Blue 'beans' product image sent by Paul to Stuart on 15 February 2023 as attachment confirming product type for Warwick drops [A5].

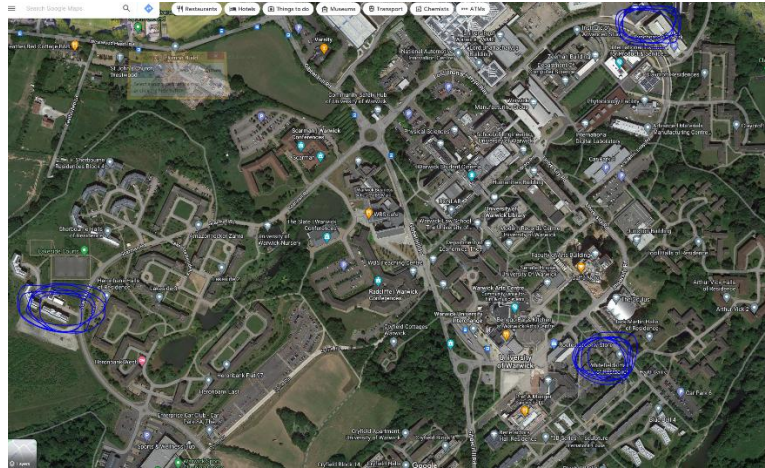


Figure A4.3 - 3drops160223.png, Annotated Warwick campus map showing three sequential drop locations for 16 February 2023, with route planning and timing, sent as email attachment (3drops.png) from Paul to Stuart [A5].



Figure A4.4 - warwick drop 190223.png



Figure A4.5 - warwick drop 230223.png

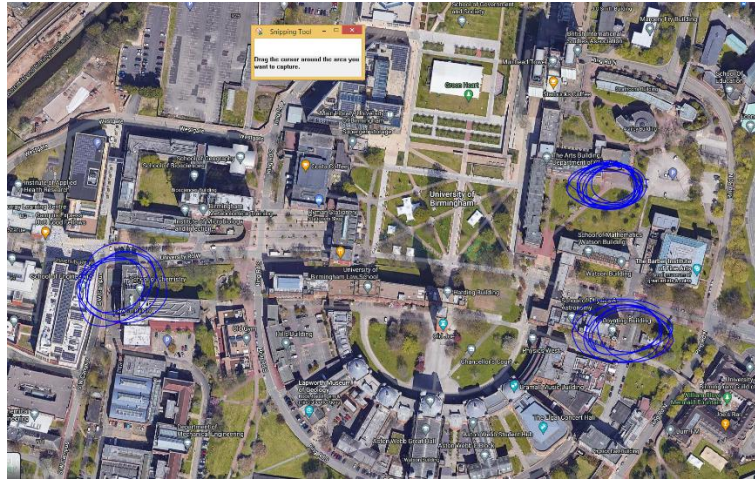


Figure A4.6 - dropoffbham.png, Annotated Birmingham University map showing collection point (Vincent Drive) and three drop locations (Arts Building, Poynting Building, School of Chemistry) with timings for 16 February 2023, sent to Drops Birmingham [A5].

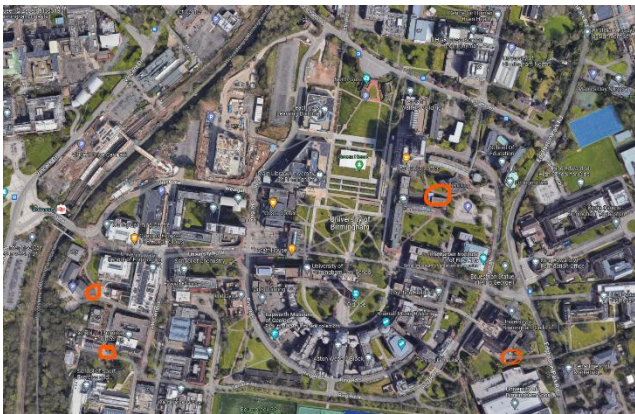


Figure A4.7 - bham drop 190223.png



Figure A4.8 - bham drop 230223.png

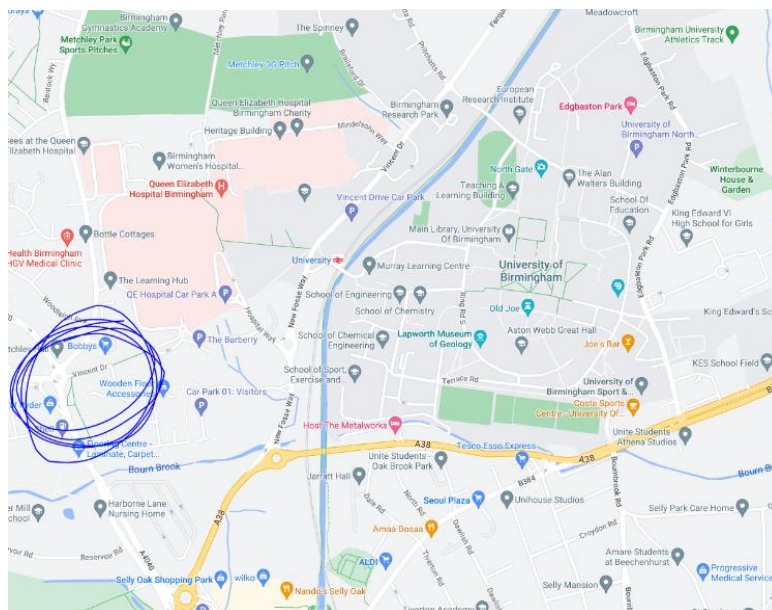


Figure A4.9 - collectionBirminghamVincentDrive.png, Google Maps view of Vincent Drive collection point for Birmingham drops, showing red Ford Focus location referenced in Paul-Drops Birmingham email instructions [A5].

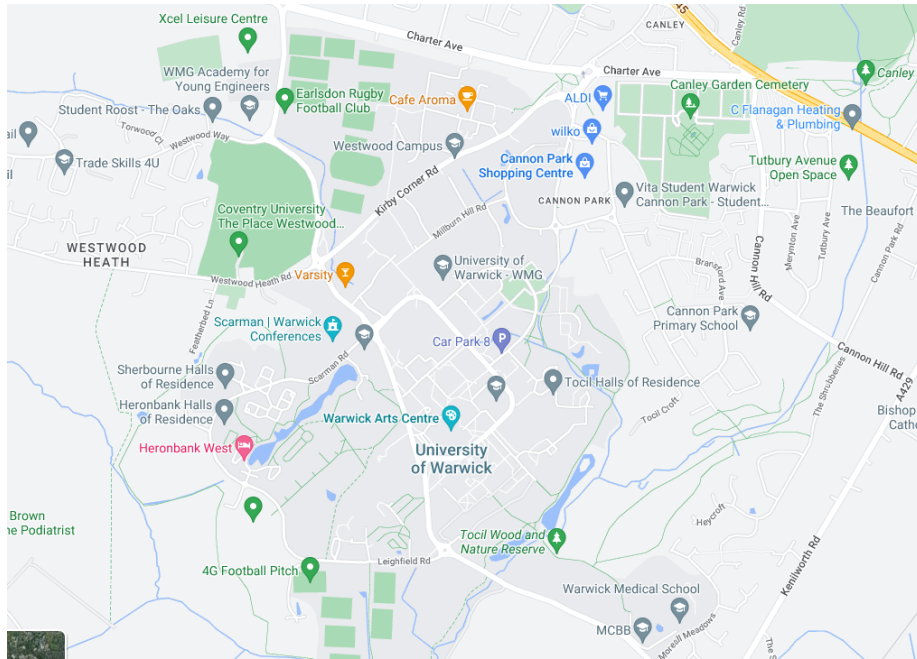


Figure A4.10 - warwick campus.png, Warwick University campus overview map accessed 16 February 2023 during drop planning period, showing familiarity with campus layout for route optimization [A8.1]

A5. Emails - Summarized Extracts

Emails between pauldjanko@gmail.com and stuarttchaimkomsiva@gmail.com

Timestamp	Direction	Content / details
14 Feb 2023 19:43	Stuart -> Paul	Introduces himself, says he has Paul's details for a university "drop", and asks Paul to confirm the date, product, price, and location.
14 Feb 2023 20:04	Paul -> Stuart	Asks for more information, asks whether Stuart has done Warwick before, and mentions there is lots of CCTV and high security.
14 Feb 2023 20:07	Stuart -> Paul	Says he knows Warwick well and has done a few drops there recently.
14 Feb 2023 20:15	Paul -> Stuart	States he does not deal with students and asks directly if Stuart is a student.
14 Feb 2023 20:19	Stuart -> Paul	Says he is not a student, explains that he covers the campus, collects from Paul's supplier, does the drop, and transfers the money.
15 Feb 2023 06:46	Paul -> Stuart	Sets out a job for "three drops on 16th": how to identify the students, a code phrase, a route to avoid CCTV, where to collect supplies (Squires Way, red Ford Focus), and cash terms of £75 per bag.
15 Feb 2023 08:50	Stuart -> Paul	Asks what his cut is, how he will be paid, what "class" the drops are, and asks where "Squire road" is (misnaming the address).
15 Feb 2023 20:12	Paul -> Stuart	Corrects the address to Squires Close, says he normally pays £30 per drop, offers to pay cash but only if they meet in Birmingham/near the airport, and includes attachment sw.png.
15 Feb 2023 20:19	Stuart -> Paul	Insists that payment must go to his bank account, says he does not want to meet in person, and again asks what "class" the drops are.

15 Feb 2023 20:22	Paul -> Stuart	Asks for Stuart's phone number, saying it is easier to chat on the phone.
15 Feb 2023 20:23	Stuart -> Paul	Refuses to share his phone number and repeats "What class.....?????" to push for clarification.
15 Feb 2023 20:26	Paul -> Stuart	Sends an attachment showing a blue beans picture (image of the product).
15 Feb 2023 20:29	Stuart -> Paul	Says £30 is too low, says blue bean drops are £40 and are risky, referring to a possible two-year sentence, and asks for the times and locations if Paul agrees to the higher rate.
15 Feb 2023 20:46	Paul -> Stuart	Agrees to £40 "on this occasion", says they need to talk about margin, and gives a detailed schedule: collect at 10:15; three drop locations and drop times; references 3drops160223.png
15 Feb 2023 20:59	Stuart -> Paul	Confirms he will keep £120 and return £90, and asks how to get the £90 to Paul because Paul's account is "useless".
15 Feb 2023 21:01	Paul -> Stuart	Tells Stuart to go back to Squires Way, mentions the red Ford Focus again, and says "don't be long please."
16 Feb 2023 12:27	Stuart -> Paul	Reports "Hey all done", confirms he kept £120, says the cash was given to Bill, comments that Bill should park further down the road, and offers to do more jobs, mentioning Coventry University and "the college".
16 Feb 2023 14:56 (email inferred through reply thread)	Paul -> Stuart	Sends "2 more for you"; says the filenames/attachments for the maps (bhamdropdate.png) explain order and everything; gives two dates (19th and 23rd) with first drop at 10:00 and collection at 09:45; confirms £35 per drop.
17 Feb 2023 11:43	Stuart -> Paul	Says the first set of instructions were missing times, suggests starting at "school of England" at 10:00 and then every 15 minutes clockwise, complains about a late driver ("Perth ski or whatever"), mentions a request for red beans, and asks if they can be supplied and how much the reds cost.
17 Feb 2023 14:08	Paul -> Stuart	Replies that, as Stuart already knows, red beans are extremely difficult to get and very risky, says his current supplier cannot get them, but he thinks he may know where to find some and will get back to Stuart.

Emails between pauldjanko@gmail.com and birminghamdrops@gmail.com

Timestamp	Direction	Content / details
15 Feb 2023 20:37	Paul -> Drops Birmingham	Says "Hi mate, I want a drop at Birmingham University, been given your number by Kevin. Can you do it? A bit last minute, but could it be done tomorrow? P."
15 Feb 2023 20:41	Drops Birmingham -> Paul	Replies "Yeah, I can do that." confirming availability.
15 Feb 2023 20:50	Paul -> Drops Birmingham	Provides full instructions: collect at Vincent Drive, red Ford Focus, at 13:00; make three drops at Arts Building at 13:30, Poynting Building at 13:45, and School of Chemistry at 14:00; describes the contact (phone in left hand, satchel, will ask for directions to

		law school); states the product is blue beans; offers £35 cash; says he can meet at Westgate; asks “is this good?”; attaches dropoffbham.png.
15 Feb 2023 20:54	Drops Birmingham -> Paul	Responds “That’s good, consider it done. How will I recognise you at Westgate?” accepting the job and asking how to identify Paul.
15 Feb 2023 20:57	Paul -> Drops Birmingham	Describes his vehicle: “white M5, last three letters SVP. will be waiting at 14:15”, giving car type, partial registration, location (Westgate), and time.
16 Feb 2023 14:09	Drops Birmingham -> Paul	Sends follow-up after the job: “Good doing business with you. let me know if you want any more done.” confirming completion and openness to more work.
16 Feb 2023 14:56	Paul -> Drops Birmingham	Sends “2 more for you”; mentions maps attached (bhamdrop190223.png, bhamdrop230223.png); gives two dates (19th and 23rd) with first drop at 10:00 and collection at 09:45; confirms price £35 per drop; mentions “lez” in the message text.

Emails about data concealment between dinnydjanko@gmail.com and reubenbeethohmahler@gmail.com

Timestamp	Direction	Content / details
17 Feb 2023 14:25	Denise Djanko -> Reuben Beethohmahler	Dinny forwards a message titled “files” and apologises for previously mis-typing Reuben’s email address. She asks for advice on the “best method of hiding files”, saying she is worried the files might be seen by someone, and explains that although she has downloaded manuals about encrypting the hard disk, they have confused her and she is concerned about “breaking something”.

A6. Registry Findings

ValuesTimeZoneInformation

Drag a column header here to group by that column

Value Name	Value Data	Value Data Raw
⌵⌵c	⌵c	⌵c
DaylightBias	-60	4294967236
DaylightName	@tzres.dll,-261	@tzres.dll,-261
StandardStart	Month 10, week of month 5, day of week 0, Hours:Minutes:Seconds:Milliseconds 2:0:0:0	00-00-0A-00-05-00-02-00-00-00-00-00-00-00-00
StandardBias	0	0
StandardName	@tzres.dll,-262	@tzres.dll,-262
Bias	0	0
DaylightStart	Month 3, week of month 5, day of week 0, Hours:Minutes:Seconds:Milliseconds 1:0:0:0	00-00-03-00-05-00-01-00-00-00-00-00-00-00-00
TimeZoneKeyName	GMT Standard Time	GMT Standard Time
ActiveTimeBias	0	0

Figure 2 - Timezone

Figure 7 - Registry USBSTOR entries indicating removable storage device connections prior to seizure, suggesting potential use of external media for data transfer or storage during investigation period.

Figure 8 - Malware under downloads

Figure 9 - Recent Documents accessed by 'PaulD'

Figure 10 - Recent documents accessed by 'djanko'

A7. Jumplist, Shellbag and Prefetch Artefacts

Name	Entry #	Target Created On	Target Modified On	Target Accessed On	Absolute Path	Extra
c9533998e1308d73.automaticDestinations-ms	0003 - 3Drops16-2-2...					
	0002 - SW.PNG					
	0001 - This PC\c:\dr...					
Properties	AppId c9533998e1308d73					
AppId d...						

Name	Entry Number	Target Created On	Target Modified On	Target Accessed On	Absolute Path	Extra
f3c90dec42360729.automaticDestinations-ms	0001 - bhamdrop 170223.pdf					
	1	2023-02-16 14:20:01	2023-02-16 14:20:01	2023-02-16 14:20:01	bhamdrop 170223.pdf	

Figure 11 - Jumplists under PaulD

```

Command line: -f D:\_MyCases\MMX-Djanko\pre\119422-MICROSOFT OFFICE 2007 SERVICE-5A44F9C0.pf
Warning: Administrator privileges not found!

Keywords: temp, tmp

Processing D:\_MyCases\MMX-Djanko\pre\119422-MICROSOFT OFFICE 2007 SERVICE-5A44F9C0.pf

Created on: 2025-12-12 18:34:04
Modified on: 2025-12-12 18:34:04
Last accessed on: 2025-12-12 18:55:44

Executable name: MICROSOFT OFFICE 2007 SERVICE
Hash: 5A44F9C0
File size (bytes): 26,618
Version: Windows 8.0, Windows 8.1, or Windows Server 2012(R2)

Run count: 2
Last run: 2023-02-16 14:43:12
Other run times: 2023-02-16 14:42:19

Volume information:
#0: Name: \DEVICE\HARDDISKVOLUME2 Serial: 2CF8F650 Created: 2023-02-12 00:28:46 Directories: 23 File references: 72
96: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\DOWNLOADS\MICROSOFT OFFICE 2007 SERVICE PACK 2.EXE
97: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\ADVAPI32.DLL
98: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\COMCTL32.DLL
99: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\GDI32.DLL
10: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\OLE32.DLL
11: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SHELL32.DLL
12: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\USER32.DLL
13: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\VERSION.DLL
14: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\USER32.DLL
15: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SECURITY.DLL
16: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\RPCRT4.DLL
17: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\COMBASE.DLL
18: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SHLWAPI.DLL
19: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\IMM32.DLL
20: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\USER32.DLL
21: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\OLEAUT32.DLL
22: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\RPCSS.DLL
23: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\USER32.DLL
24: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\CRYPTBASE.DLL
25: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\BCRYPTPRIMITIVE.DLL
26: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\UXTHEME.DLL
27: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SHFOLDER.DLL
28: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SHCORE.DLL
29: \DEVICE\HARDDISKVOLUME2\WINDOWS\GLOBALIZATION\SORTING\SORTDEFAULT.NLS
30: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\SETUPAPI.DLL
31: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\CFGMGR32.DLL
32: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\EN-US\SETUPAPI.DLL.NUI
33: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\CLBCATQ.DLL
34: \DEVICE\HARDDISKVOLUME2\WINDOWS\SYSTEM32\PROPSYS.DLL
35: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\MICROSOFT\WINDOWS\CACHES\CVERSIONS.1.DB
36: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\MICROSOFT\WINDOWS\CACHES\{AFBF9F1A-8EE8-4C77-AF34-C647E37CA0D9}.1.VER0X0000000000000000.DB
37: \DEVICE\HARDDISKVOLUME2\USERS\DESKTOP.INI
38: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\DOWNLOADS\DESKTOP.INI
39: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\DESKTOP\DESKTOP.INI
40: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\DOCUMENTS\DESKTOP.INI
41: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\MUSIC\DESKTOP.INI
42: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\PICTURES\DESKTOP.INI
43: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\VIDEOS\DESKTOP.INI
44: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\TEMP\MSI7603.TMP (Keyword: True)
45: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\TEMP\MSI7604.TMP\SYSTEM.DLL (Keyword: True)
46: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\TEMP\MSI7604.TMP\WORTHSTAR.EXE (Keyword: True)
47: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\TEMP\MSR8514.TMP (Keyword: True)
48: \DEVICE\HARDDISKVOLUME2\USERS\DJANKO\APPDATA\LOCAL\TEMP\MSR8515.TMP\SYSTEM.DLL (Keyword: True)

```

Figure 12 - Prefetch file demonstrating 'microsoft office 2007 service pack 2.exe' was executed twice on 16 February 2023, with run count and last execution timestamp supporting temporal analysis of potential malware activity [A9].

Source Name	Path	Key	Last Write	Data Source	Date Modified	Date Created	Date Accessed
NTUSER.DAT	Recycle Bin	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01			
NTUSER.DAT	3Drops16-2-23.PNG	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 06:54:02 GMT	2023-02-15 06:54:02 GMT	2023-02-15 06:54:02 GMT
NTUSER.DAT	blue.png	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 20:25:36 GMT	2023-02-15 20:25:36 GMT	2023-02-15 20:25:36 GMT
NTUSER.DAT	colletionBirmingham	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 20:39:02 GMT	2023-02-15 20:39:02 GMT	2023-02-15 20:39:02 GMT
NTUSER.DAT	dropoffbham.PNG	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 20:40:28 GMT	2023-02-15 20:40:28 GMT	2023-02-15 20:40:28 GMT
NTUSER.DAT	SW.PNG	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 06:52:08 GMT	2023-02-15 06:52:08 GMT	2023-02-15 06:52:08 GMT
NTUSER.DAT	warwick campus.PNG	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-15 06:51:06 GMT	2023-02-15 06:51:06 GMT	2023-02-15 06:51:06 GMT
NTUSER.DAT	bhamdrop 170223.pdf	Software\Microsoft\Windows\Shell\Bi	2023-02-16 14:51:33 GMT	MMX-Djanko-PC.E01	2023-02-16 14:20:02 GMT	2023-02-16 14:20:02 GMT	2023-02-16 14:20:02 GMT
UsrClass.dat	My Computer\C:\User\Paul\Download	Local Settings\Software\Microsoft\Win	2023-02-16 14:23:37 GMT	MMX-Djanko-PC.E01	2023-02-16 14:23:34 GMT	2023-02-14 19:18:00 GMT	2023-02-16 14:23:34 GMT

Figure 13 - Shellbag indicating user PaulD's access

A8. Browser Artefacts

URL	Date Accessed	Referrer URL	Title	Program & Domain	Data Source
https://www.dropbox.com/sh/q60tq401nrfcv/AAB6i4MTN5Uicx	2023-02-16 12:29:28 GMT	https://www.dropbox.com	Dropbox	Google Chrome	MMX-Djanko-PC.E01
https://www.dropbox.com/profile_services/start_auth_flow?factio	2023-02-16 12:28:14 GMT	https://www.dropbox.com	Dropbox	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/search?q=how+to+hide+files+on+windo	2023-02-17 14:26:39 GMT	https://www.google.com	Google Search	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/search?q=how+to+encrypt+a+partitio	2023-02-17 14:26:40 GMT	https://www.google.com	Google Search	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/maps/@52.462612,-1.9530504,13z	2023-02-16 14:07:02 GMT	https://www.google.com	Google Maps	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/maps/@52.453551,-1.9306978,16z	2023-02-16 14:07:17 GMT	https://www.google.com	Google Maps	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/maps/@52.4506186,-1.9301627,232m/data	2023-02-16 14:08:36 GMT	https://www.google.com	Google Maps	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/maps/@52.4498748,-1.927978,465m/data	2023-02-16 14:10:46 GMT	https://www.google.com	Google Maps	Google Chrome	MMX-Djanko-PC.E01
https://download.cnet.com/PDF-Reader-for-Windows-4/3000-1885	2023-02-16 14:22:17 GMT	https://download.cnet.com	Free download and software reviews - CNET Download	Google Chrome	MMX-Djanko-PC.E01
https://erfssoft.ru/pdf-reader/702-BIIBH8BDAOTD1RRAJFTVvYNYC	2023-02-16 14:22:55 GMT	https://erfssoft.ru/	PDF reader Free Download	Google Chrome	MMX-Djanko-PC.E01
https://alt.sharplifedownload.com/d/OBOD0IB?title=Pdf+reader	2023-02-16 14:23:24 GMT	https://alt.sharplifedownload.com	Pdf reader installer	Google Chrome	MMX-Djanko-PC.E01
https://floppydirect.click/7jy007e1a2509e8f1480afac34651189d	2023-02-16 14:23:25 GMT	https://floppydirect.click	Pdf reader installer	Google Chrome	MMX-Djanko-PC.E01
https://s3.us-east-2.amazonaws.com/zcomajuly/xfig136bzr/Pdfre	2023-02-16 14:23:31 GMT	https://s3.us-east-2.amazonaws.com	Pdf reader installer	Google Chrome	MMX-Djanko-PC.E01
https://www.avast.com/dk-dark-web	2023-02-17 14:00:11 GMT	https://www.avast.com	What is the Dark Web & How to Access It?	Google Chrome	MMX-Djanko-PC.E01
https://www.compartech.com/blog/vpn-privacy/access-dark-we	2023-02-17 14:00:24 GMT	https://www.compartech.com	How to Access the Dark Web safely: Updated Guide for 2023	Google Chrome	MMX-Djanko-PC.E01
https://www.kaspersky.com/resource-center/threats/deep-we	2023-02-17 14:00:31 GMT	https://www.kaspersky.com	What is the Dark Web Dangerous? What you need to know	Google Chrome	MMX-Djanko-PC.E01
https://sopa.tulane.edu/blog/everything-you-should-know-abou	2023-02-17 14:00:24 GMT	https://sopa.tulane.edu	Everything You Should Know About the Dark Web Tulane School of Professional Advancement	Google Chrome	MMX-Djanko-PC.E01
https://www.thinkuknow.co.uk/professionals/our-views/the-d	2023-02-17 14:00:25 GMT	https://www.thinkuknow.co.uk	The Dark Web: What is it and why do people use it?	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/search?q=how+to+access+the+dark+we	2023-02-17 14:00:28 GMT	https://www.google.com	Google Search	Google Chrome	MMX-Djanko-PC.E01
https://www.kaspersky.com/resource-center/threats/deep-we	2023-02-17 14:00:31 GMT	https://www.kaspersky.com	What is the Dark Web Dangerous? What you need to know	Google Chrome	MMX-Djanko-PC.E01
https://www.zdnet.com/article/your-complete-guide-to-the	2023-02-17 14:01:48 GMT	https://www.zdnet.com	Your guide to the dark web and how to safely access .onion websites ZDNET	Google Chrome	MMX-Djanko-PC.E01
https://clarico.co/blog/is-dark-web-illegal/	2023-02-17 14:00:34 GMT	https://clarico.co	Is The Dark Web Illegal? How To Access It Safely?	Google Chrome	MMX-Djanko-PC.E01
https://www.aura.com/learn/dark-web	2023-02-17 14:00:42 GMT	https://www.aura.com	What is the Dark Web? How It Works & Why It's So Dangerous Aura	Google Chrome	MMX-Djanko-PC.E01
https://www.imf.org/en/Publications/fandd/issues/2019/09/th	2023-02-17 14:01:18 GMT	https://www.imf.org	the-truth-about-the-dark-web-kumar	Google Chrome	MMX-Djanko-PC.E01
https://www.expressvpn.com/blog/how-to-access-the-dark-we	2023-02-17 14:01:31 GMT	https://www.expressvpn.com	ExpressVPN Blog	Google Chrome	MMX-Djanko-PC.E01
https://setapp.com/how-to/get-on-dark-web	2023-02-17 14:01:31 GMT	https://setapp.com	How to get on the dark web	Google Chrome	MMX-Djanko-PC.E01
https://www.childrensociety.org.uk/information/profession	2023-02-17 14:01:31 GMT	https://www.childrensociety.org.uk	The Children's Society	Google Chrome	MMX-Djanko-PC.E01
https://apps.apple.com/ie/app/private-browser-the-dark-we	2023-02-17 14:02:12 GMT	https://apps.apple.com	Private Browser: The Dark Web on the App Store	Google Chrome	MMX-Djanko-PC.E01
https://www.bulkwholesalesweets.co.uk/collections/wholes	2023-02-17 14:02:34 GMT	https://www.bulkwholesalesweets.co.uk	Bulk Wholesale Jelly Beans	Google Chrome	MMX-Djanko-PC.E01
https://www.google.com/search?q=Supplier+red+jelly+beans	2023-02-17 14:02:36 GMT	https://www.google.com	Supplier red jelly beans Birmingham - Google Search	Google Chrome	MMX-Djanko-PC.E01
https://www.candyroom.co.uk/jelly-beans-cherry/	2023-02-17 14:02:57 GMT	https://www.candyroom.co.uk	Candy Jelly Beans Very Cherry - Candy Room	Google Chrome	MMX-Djanko-PC.E01

Figure A8.1 - Browser history entries showing Google searches, dark-web access guidance pages, and visits to the Candy Room "jelly beans" product page on 17 February 2023, from artefacts recovered on MMX-Djanko-PC.E01.

Source Na	URL	Domain	Date Created	Headers	Path	Data Source
data_1	1/0/_dk_https://avast.com/https://avast.com/https://academy.avast.co	avast.com	2023-02-17 14:01:30 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://tulane.edu/https://tulane.edu/https://sopa.tulane.edu	tulane.edu	2023-02-17 14:00:25 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://apple.com/https://apple.com/https://s4-ssl.mzstatic.c	mzstatic.com	2023-02-17 14:02:05 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://setapp.com/https://setapp.com/https://cdn.setapp.co	setapp.com	2023-02-17 14:02:01 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://expressvpn.com/https://expressvpn.com/https://sp-ac	pcdn.co	2023-02-17 14:02:43 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://candyroom.co.uk/https://candyroom.co.uk/https://js.s	stripe.com	2023-02-17 14:03:50 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://clarico.co/https://clarico.co/https://static-cdn.clarico	co/bclarico	2023-02-17 14:00:35 GMT	date: Mon, 13 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://childrensociety.org.uk/https://childrensociety.org	uk/childrensociety.org	2023-02-17 14:02:41 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://all8soft.com/https://all8soft.com/https://all8soft.c	om/all8soft.com	2023-02-16 14:02:41 GMT	date: Thu, 16 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://bulkwholesalesweets.co.uk/https://bulkwholesalesw	bulkwholesalesweets.co.uk	2023-02-17 14:02:51 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://compartech.com/https://compartech.com/https://cor	content-autofill.googleapis	2023-02-17 14:01:37 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://candyroom.co.uk/https://candyroom.co.uk/https://ww	candyroom.co.uk	2023-02-17 14:03:42 GMT	date: Fri, 17 Feb 2023	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01
data_1	1/0/_dk_https://apple.com/https://apple.com/https://sp.apple.com/co	apple.com	2023-02-17 14:02:49 GMT	access-control-allow-	/img_MMX-Djanko-PC.E01/vol_vol3/Users,MMX-Djanko-PC.E01	MMX-Djanko-PC.E01

Figure A8.2 - Browser cache entries linking Candyroom.co.uk pages to embedded third-party resources including Stripe fingerprinting scripts, Shopify, and other content loaded during site access on 17 February 2023, MMX-Djanko-PC.E01.

Source Na	URL	Date Accessed	Name	Program Name	Domain	Data Source
Cookies	floppyredirect.click	2023-02-16 14:23:24 GMT	mss	Google Chrome	floppyredirect.cl	MMX-Djanko-PC.E01
Cookies	.tulane.edu	2023-02-17 14:17:15 GMT	_hjSession_2245180	Google Chrome	tulane.edu	MMX-Djanko-PC.E01
Cookies	www.candyroom.co.uk	2023-02-17 14:04:05 GMT	mailchimp_landing_site	Google Chrome	www.candyroor	MMX-Djanko-PC.E01
Cookies	www.bulkwholesalesweet	2023-02-17 14:02:50 GMT	shopify_pay_redirect	Google Chrome	www.bulkwhole	MMX-Djanko-PC.E01
Cookies	www.candyroom.co.uk	2023-02-17 14:04:05 GMT	woocommerce_recently_viewed	Google Chrome	www.candyroor	MMX-Djanko-PC.E01
Cookies	.expressvpn.com	2023-02-17 14:03:42 GMT	_ga	Google Chrome	expressvpn.com	MMX-Djanko-PC.E01
Cookies	.bulkwholesalesweets.co.uk	2023-02-17 14:03:40 GMT	_shopify_s	Google Chrome	bulkwholesalesv	MMX-Djanko-PC.E01
Cookies	.bulkwholesalesweets.co.uk	2023-02-17 14:03:40 GMT	_shopify_y	Google Chrome	bulkwholesalesv	MMX-Djanko-PC.E01
Cookies	www.bulkwholesalesweet	2023-02-17 14:03:40 GMT	secure_customer_sig	Google Chrome	www.bulkwhole	MMX-Djanko-PC.E01
Cookies	.www.candyroom.co.uk	2023-02-17 14:04:05 GMT	_stripe_mid	Google Chrome	www.candyroor	MMX-Djanko-PC.E01
Cookies	.www.candyroom.co.uk	2023-02-17 14:04:05 GMT	_stripe_sid	Google Chrome	www.candyroor	MMX-Djanko-PC.E01
Cookies	.candyroom.co.uk	2023-02-17 14:03:54 GMT	_gid	Google Chrome	candyroom.co.u	MMX-Djanko-PC.E01
Cookies	m.stripe.com	2023-02-17 14:04:05 GMT	m	Google Chrome	m.stripe.com	MMX-Djanko-PC.E01

Figure A8.3 - HTTP cookies created while accessing Candyroom.co.uk and related commerce infrastructure (Mailchimp, Shopify, Stripe) at around 14:04 GMT on 17 February 2023, showing persistence of session and payment-flow data on MMX-Djanko-PC.E01.

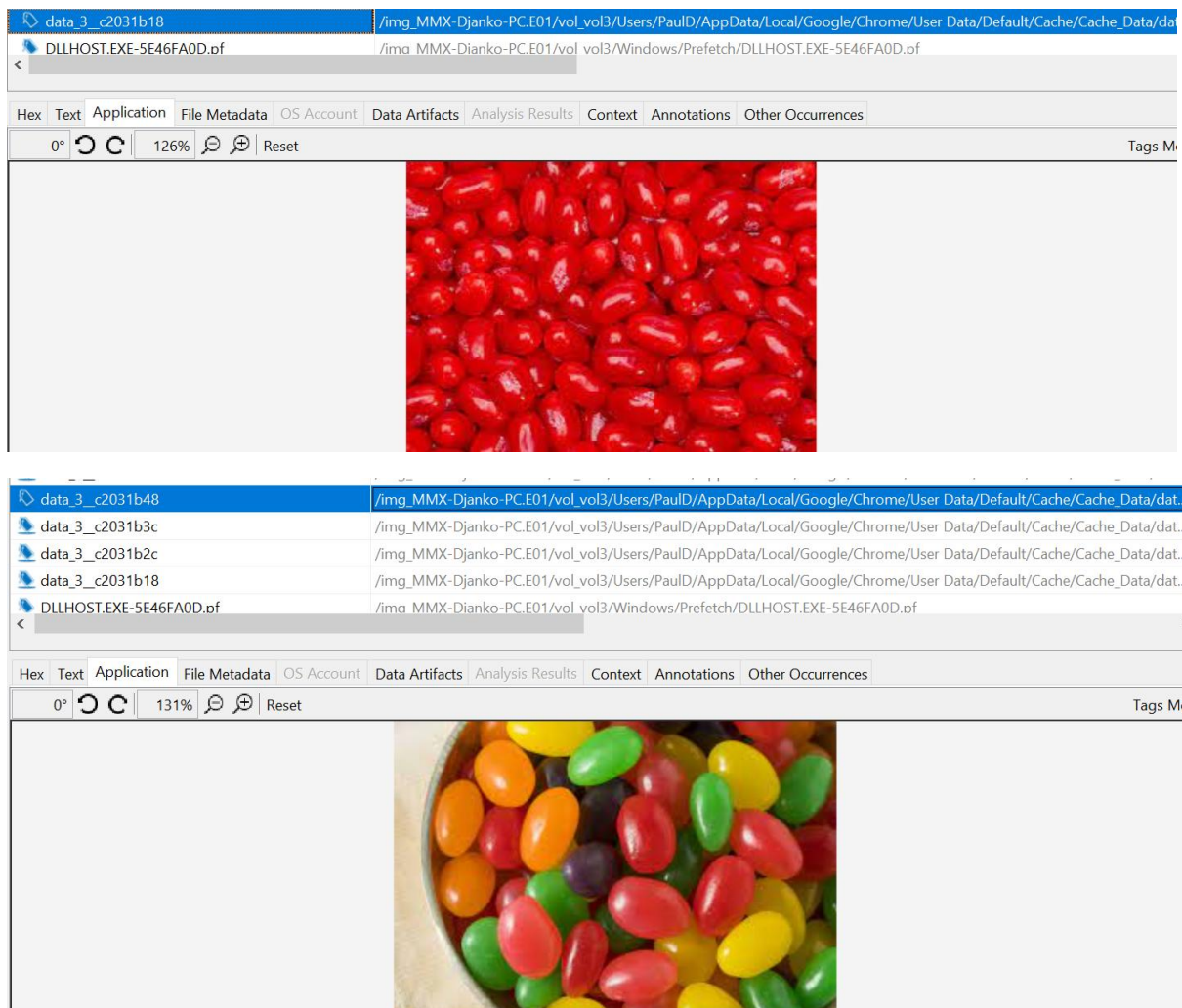


Figure 14 - In the context of Paul's earlier searches for 'supplier red jelly beans Birmingham' and email discussions about obtaining 'red beans', these cached product images supports an inference that he was actively looking at potential red-bean as well as mixed bean suppliers.

A9. Potential Executables

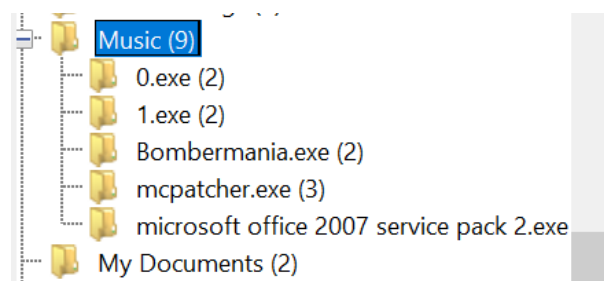


Figure 15 - PaulD's Music Folder containing malware folders

StartSetup_20221.exe		0	File	Notable		Rules
StartSetup_20221.exe		0	File	Notable		Rules
StartSetup_20221.exe		0	File	Notable		Rules
StartSetup_20221.exe		0	File	Notable		Rules
mcpatcher.exe		0	File	Notable		Rules
microsoft office 2007 service pack 2.exe		0	File	Notable		Rules

Figure 16 - Executables marked by YARA rule hits

The figure displays three VirusTotal analysis results. The top two results are for 'mcpatcher.exe' and 'microsoft office 2007 service pack 2.exe', both showing a community score of 62/72 and being flagged as malicious by 43/72 security vendors. The bottom result is for '7zS.sfx.exe', showing a community score of 22/72 and being flagged as adware by 22/72 security vendors. Each result includes a 'Security vendors' analysis table with various security companies and their verdicts.

Vendor	Verdict
AbrLab-V3	Win-PUP/Solima.X1373
ALYac	Gen-Variant.Adware.Solima.1
Arctic Wolf	Unsafe
AVG	Win32-PUP-gen (PUP)
Avira (no cloud)	Adware (no cloud)
Avira (no cloud)	PUP/Solima.Gen
Avira (no cloud)	Adware (no cloud)
Avira (no cloud)	PUP/Solima.Gen
Avira (no cloud)	Adware (no cloud)
Avira (no cloud)	PUP/Solima.Gen

Vendor	Verdict
Bkav Pro	W32.AIDetectMalware
DeepInstinct	MALICIOUS
ESET-NOD32	Win32/Adware.OpenUpdater.KE Applic...
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1
Gridinsoft (no cloud)	Adware.Win32.DownWare.oais1

Figure 17 - mcpatcher.exe, microsoft office pack 2008.exe, StartSetup_20221.exe flagged as adware on VirusTotal